* Inferential STATS : It is a part of statistics where we take a small sample, analyze it and make conclusions or predictions about the larger popul^n.
* Standardization ⇒ Z score ⇒ We can use in ML and also in time series.

In machine learning zscore is called standardization.
In Time series zscore is called white noise.

$$\boxed{\text{Z-score} = \frac{x - \mu}{\sigma}}$$

Z score tells us how far a value is from the mean in terms of standard deviations

Original dataset

1
2
2
3
3
3
4
4
5

Mean = 3
Std dev = 1.22

$x - \mu$

-2
-1
-1
0
0
0
1
1
2

Mean = 0
Std dev = 1.22

$\frac{x-\mu}{\sigma}$

-1.63

-0.81

-0.81

0

0 mean = 0

0 std = 1

0.81

0.81

1.63

Hence z-score is a standardization technique used to convert mean to 0 and standard deviation to 1

* Confidence Interval : A confidence level tells you how sure you are that your sample results represent ~~the~~ whole population.

95% confident in math exam score b/w 90-95

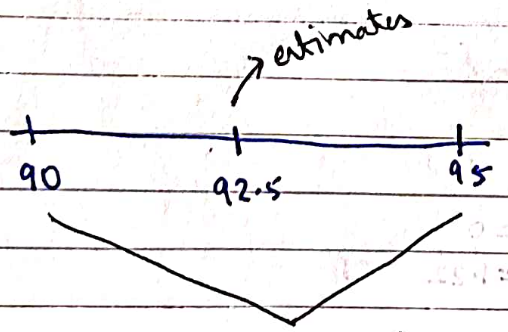
99% confident in math exam will score b/w 92-98

error = 5% (alpha) ⇒ p-value (The error term is called alpha)

5% = 5/100 = 0.05 considered as p-value.

97 -- 99% test cases are passed

92 -- 95% test cases are passed.



These two points are called estimator

The lower and upper bound of confidence interval is called estimator

The mean of the estimator is called estimates.

Confidence interval is divided into 2 parts

population variance known	population variance unknown
(z-test)	(t-test)
Used when sample is big (for population)	Used when the sample is small (for sample)

* Advanced Statistics

① Hypothesis testing

statement -- Hyd kiwi fruit is expensive (Just a sentence)

1 kiwi fruit is 2000!!! -- Hypothesis. (A statement which you want to test if its true or not)

Hypothesis
- Null Hypothesis (H0)
- Alternative Hypothesis (H1 or HA)

Null Hypothesis (H_0)
The null hypothesis says there is no difference. Nothing new is happening.

Alternative Hypothesis (H_1 or H_a)
The alternative hypothesis says there is a difference or effect. Something new is happening.

① accept the null hypothesis
② reject the null hypothesis | accept the alternative hypothesis.

$\boxed{\text{Far from mean} \Rightarrow \text{reject the null hypothesis}}$